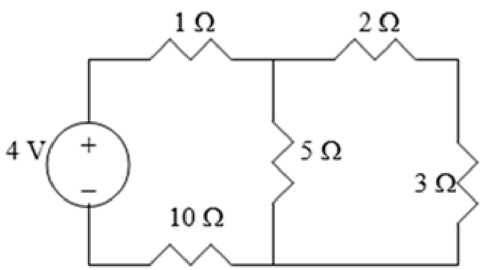
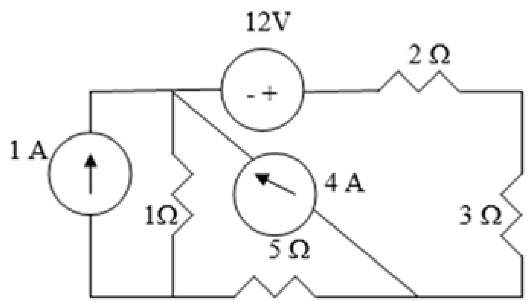
ENGR 2303: Electrical Science

HW 2_Ch2

1) Find the number of branches and nodes in each of the circuits below

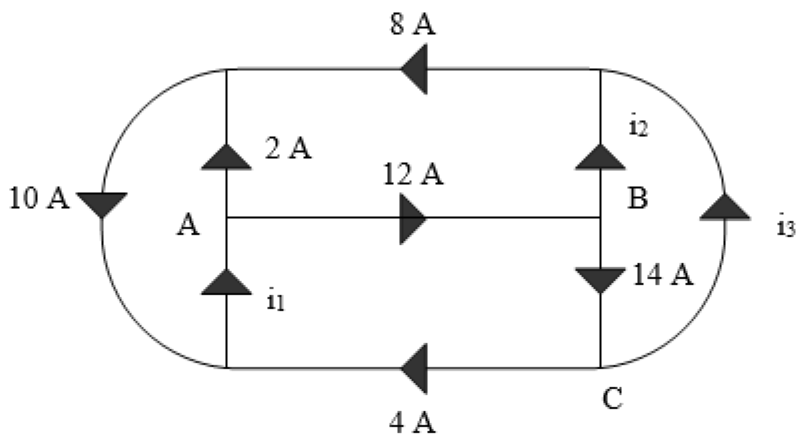


(a)

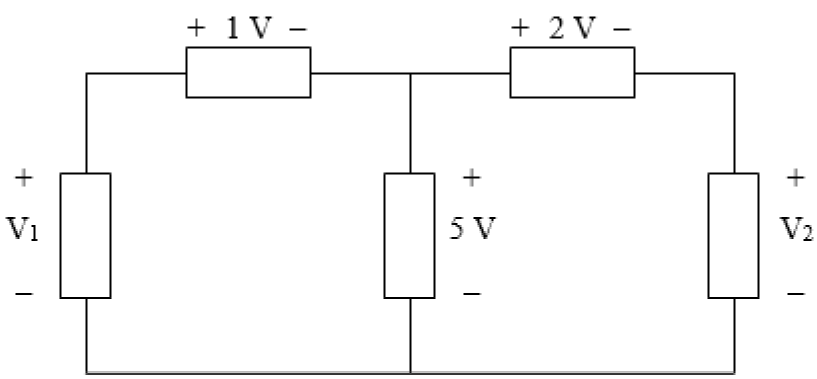


(b)

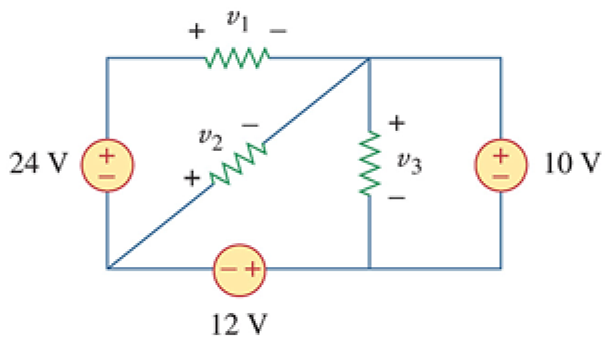
2) Find i_1, i_2, and i_3



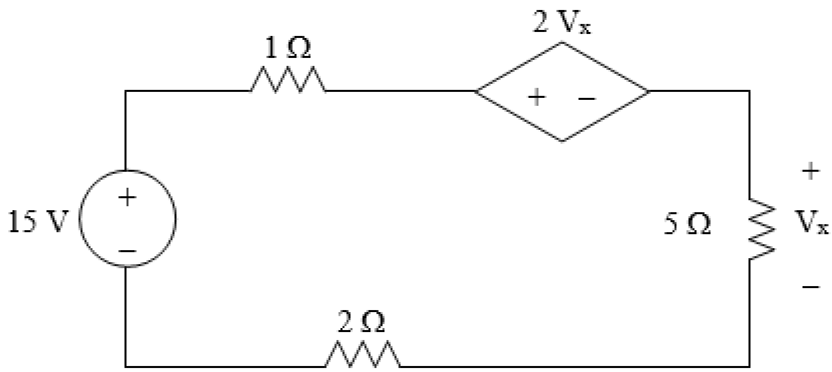
3) In the circuit below, calculate V_1 and V_2.



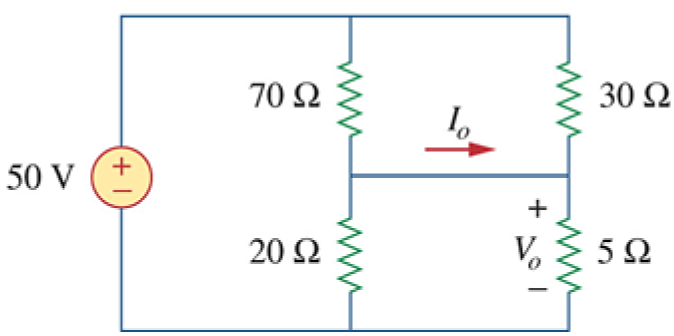
4) Obtain v_1, v_2, and v_3 in the circuit below



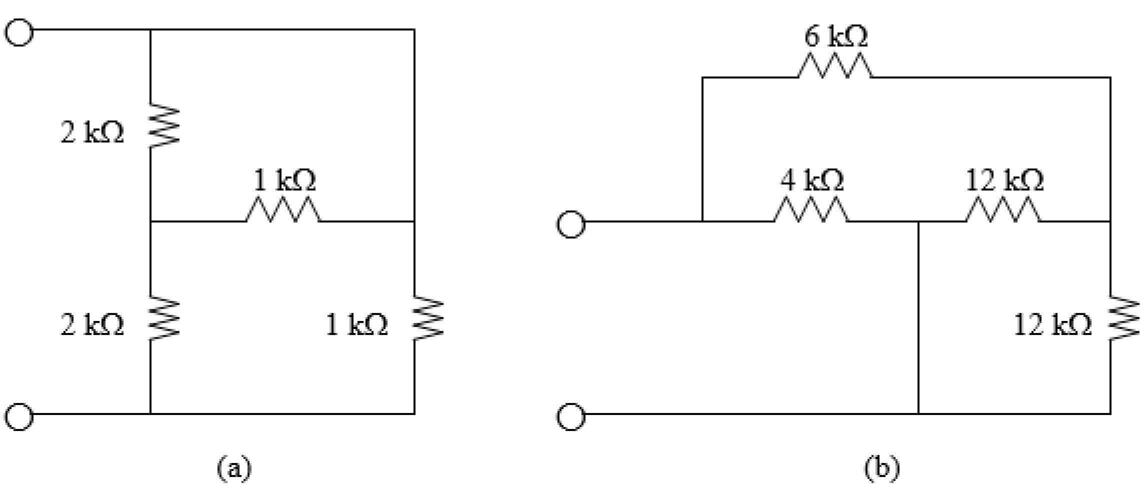
5) Find V_x in the circuit below



6) Calculate V_o and I_o in the circuit below



7) Evaluate R_{eq} for each of the circuits shown below



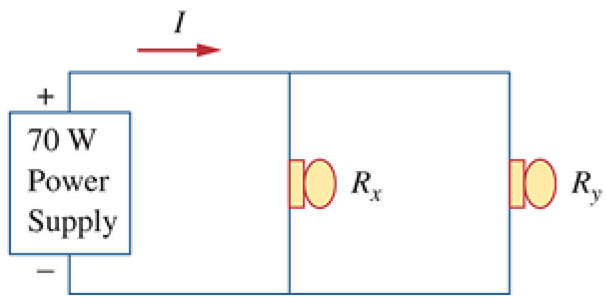
8) As a design engineer, you are asked to design a lighting system consisting of a 70-W power supply and two lightbulbs as shown in figure below. You must select the two bulbs from the following three available bulbs.

R_1 = 80Ω, cost = \$0.60 (standard size)

R_2 = 90Ω, cost = \$0.90 (standard size)

R_3 = 100 Ω, cost = \$0.75 (nonstandard size)

The system should be designed for minimum cost such that I = 1.2 A $\pm$ 5 percent.



9) Two delicate devices are rated as shown in figure below. Find the values of the resistors R_1 and R_2 needed to power the devices using a 24-V battery.